

Statistical Arbitrage in Crypto Pair Spreads: Kalman Hedging, Selection Bias, and the Limits of Predictability

Phase 1 Report

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Abstract

We study statistical arbitrage on hourly Binance.US USDT spot pairs, adapting the equity pairs-trading framework of Gatev et al. (2006) to cryptocurrency markets. The central methodological finding is that a static ordinary-least-squares (OLS) hedge ratio fails out-of-sample on this market: of 1,219 candidate pairs in the top-50 liquidity universe at $t_0 = 2024-01-01$, only 5.4% maintain an augmented Dickey–Fuller (ADF) $p < 0.05$ on a held-out 30-day test slice when the hedge ratio is fit only on the preceding 90-day training window. A Kalman filter with parameters maximum-likelihood-fit on training data and forward-rolled on test recovers cointegration on 99.7% of the same pairs at the same threshold. The relationships in crypto pairs are real but non-stationary; a fixed hedge ratio is the wrong model.

We further evaluate four classifiers (a gradient-boosted tree, an LSTM, and two transformers of different size) on a 27-split walk-forward protocol across the Kalman-derived spreads. The classifiers’ predictive signal collapses, in a one-feature-out ablation, almost entirely to the single feature spread_z, which is the same quantity the classical z -threshold rule already uses. After a Bailey–López de Prado deflated-Sharpe correction for the $N \approx 50$ strategy configurations evaluated in this project, the pre-cost annualized Sharpe of ~ 8 fails to reject the null of zero true Sharpe at realistic trial counts; at maker-fee-equivalent transaction costs the realized Sharpe drops to ~ 0.9 and sits below the chance-best of ~ 1.2 at $N = 10$. We retract the corresponding break-even claim.

The narrower defensible market-efficiency statement: at hourly cadence on Binance.US USDT pairs, the mean-reverting alpha resides in the $|\text{spread}_z| \geq 2$ entry condition itself, not in the choice of classifier. The Kalman-spread cointegration result is robust and the load-bearing finding of this phase. L2 microstructure data, expected in Phase 2, is required to test whether intra-bar execution improves the post-cost picture.

1 Introduction

Gatev et al. (2006) (henceforth GGR) propose a systematic pairs-trading rule on US equities: select pairs by historical distance, trade the spread when it exceeds 2 standard deviations from its mean,

close at convergence. Their reported Sharpe of ~ 0.5 – 1.0 has motivated decades of follow-on work in equities, commodities, and more recently digital assets.

The cryptocurrency market differs from the GGR setting in three material ways. First, the investable universe changes rapidly: new tokens are listed weekly and others are delisted, often without warning. Second, the typical relationship between two tokens is unstable on multi-month horizons; pairs that look cointegrated on one year of data may show no relationship the next. Third, exchange fees are large relative to typical spread movements: Binance.US default taker fees of 10 bps per side imply 20 bps per round-trip per leg of a pair trade.

This work adapts the GGR framework to hourly Binance.US spot data with USDT quote, with two specific contributions:

1. A direct out-of-sample comparison of static OLS hedge ratios against a Kalman-filter dynamic hedge across the full universe of 1,219 liquidity-filtered pairs, showing that the time-varying hedge recovers cointegration almost universally where the static hedge fails.
2. An honest evaluation, using block-bootstrap confidence intervals and a deflated Sharpe correction (Bailey and López de Prado, 2014), of whether modern machine-learning models provide a tradable edge on Kalman-derived spreads after transaction costs and after accounting for the multiple-testing inherent in our methodology sweep.

We find that the first contribution is unambiguous and constitutes the load-bearing finding of this phase. The second is largely negative: the ML models we tried do not produce a selection-corrected Sharpe that beats chance at realistic transaction-cost levels, and most of the “alpha” visible in our backtests is attributable to the $|z| \geq 2$ entry condition rather than the classifier that fires it.

The remainder of this paper is organized as follows. Section 2 describes the data and the as-of-time universe construction. Section 3 details the methodology, including the Kalman state-space specification, the feature set, the classifier architectures, and the evaluation protocol. Section 4 presents the four headline findings. Section 5 catalogs limitations. Section 6 describes the Phase 2 L2-microstructure plan.

2 Data

Source. Hourly open-high-low-close-volume (OHLCV) candles from the Binance.US REST endpoint `api.binance.us/api/v3/klines`, USDT quote only. The local store contains 195 currently-listed symbols, with first-observation dates ranging from 2019-09-23 (BTC, ETH, BNB, XRP, BCH) to 2025-12-18. All symbols share the same final bar at 2026-01-22 04:00 UTC, the snapshot date.

As-of-time universe. A symbol enters the universe at calendar time t_0 if and only if its first observation precedes $t_0 - \tau_{\min}$ and its last observation postdates $t_0 - \Delta t$, where $\tau_{\min}$ is a minimum-history requirement and Δt is one bar. We use $\tau_{\min} = 180$ days throughout. The first-observation check is necessary to prevent look-ahead leakage when t_0 is in the past; an earlier version of our universe function omitted this check and included symbols whose listing postdated the analysis cutoff.

Liquidity filter. Within the as-of universe at t_0 , we rank symbols by mean USDT volume (close $\times$ volume) over the training window and keep the top 50. Symbols with less than 80% coverage of the expected hourly calendar in the window are dropped before ranking. At $t_0 = 2024-01-01$, this yields a 130-symbol universe and a top-50 liquid set.

Survivorship. Because the snapshot was downloaded as a single point in time and only contains currently-listed symbols, the panel cannot observe pre-snapshot delistings. We compiled an external delisting log from the Binance.US Help Center announcements: 14 tokens were delisted between t_0 and the snapshot date.¹ This implies a survivorship rate of approximately $14/209 = 6.7\%$ as a loose upper bound. After the liquidity top-50 filter, only an estimated 3–6 of these would have been candidates; the remainder were small-cap tokens that would not have passed the liquidity threshold even at t_0 . The realistic bias on the headline metrics is therefore 5%–12% of the pre-filter candidate set.

3 Methodology

3.1 Pair selection

We screen pairs in two stages. The strict cointegration screen, adapted from GGR, requires (i) Pearson correlation of log returns ≥ 0.5 , (ii) ADF p -value on the OLS residual ≤ 0.05 , (iii) Ornstein–Uhlenbeck half-life in $[4, 240]$ hours, and (iv) beta stability (standard deviation of OLS β across four equal segments, divided by absolute mean β) ≤ 0.35 . On the local Binance.US slice at $t_0 = 2024-01-01$, this strict screen finds *zero* passing pairs. We therefore use a correlation fallback (rank by Pearson correlation of returns, take top N) for the downstream pipeline, and treat the failure of the strict screen as evidence that motivates the Kalman analysis of Section 3.2.

3.2 Static and dynamic hedge ratios

For each pair (A, B) with log prices $y_t = \log p_t^A$ and $x_t = \log p_t^B$, the *static OLS hedge* is the pair (α, β) minimizing $\sum_t (y_t - \alpha - \beta x_t)^2$ over a training window. The residual $\epsilon_t = y_t - \alpha - \beta x_t$ is the static spread.

The *Kalman dynamic hedge* treats (α, β) as a hidden state evolving as a random walk:

$$\mathbf{s}_t = \begin{bmatrix} \alpha_t \\ \beta_t \end{bmatrix}, \quad \mathbf{s}_t = \mathbf{s}_{t-1} + \mathbf{w}_t, \quad \mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, Q), \quad (1)$$

$$y_t = \begin{bmatrix} 1 & x_t \end{bmatrix} \mathbf{s}_t + v_t, \quad v_t \sim \mathcal{N}(0, R), \quad (2)$$

with $Q = \text{diag}(q_\alpha, q_\beta)$. The process noise Q and observation noise R are unknown parameters and are fit by maximum likelihood on a training slice, with the likelihood being the standard prediction-error decomposition,

$$\log L(Q, R) = -\frac{1}{2} \sum_t \left[\log(2\pi S_t) + \frac{(y_t - \hat{y}_{t|t-1})^2}{S_t} \right], \quad (3)$$

where $\hat{y}_{t|t-1}$ is the one-step-ahead prediction and S_t is its variance under the filter. We optimize $(\log q_\alpha, \log q_\beta, \log R)$ by L-BFGS-B; the log parameterization keeps the variances positive.

After fitting, we run the filter forward once more on the training slice with the fitted parameters, record the final filtered state $\mathbf{s}_{T_{\text{train}}}$ and the training-period residuals $\hat{\epsilon}_t^{\text{train}}$. On the test slice we initialize the filter at $\mathbf{s}_{T_{\text{train}}}$, hold the fitted (Q, R) fixed, and produce out-of-sample residuals $\hat{\epsilon}_t^{\text{test}}$. We perform *no* re-fitting on test data. The training and test residual series are concatenated into the spread series used downstream, with the same filter parameters operative on both sides of the train/test boundary.²

¹BOND, ANT, WAVES, TUSD, CUDOS, MXC, REN, VITE, BAL, CLV, STMX, LOOM, KDA, OXT, JAM.

²An earlier version of this pipeline used different parameters on the two sides (default Kalman config for training, MLE-fit for test), which produced a discontinuity at the boundary. This was caught in audit and fixed; see Section 3.8.

3.3 Features and labels

Per pair, per hourly bar, we compute 16 features in three groups: spread features (spread , spread_z^{168h} , $\text{spread}_{\Delta}^{1h,4h,24h}$, $\text{spread}_{\sigma}^{24h,168h}$), volume features (volume ratio, volume z -scores, pair volume sum/geometric mean), and market-context features (BTC return 24h , rolling correlation 24h , realized volatility $_{A,B}^{24h}$). All features are backward-rolling. The target is the change in absolute spread over the next 24 hours,

$$\Delta_t = |s_{t+24}| - |s_t|, \quad (4)$$

quantized into three classes: 0 (revert, $\Delta_t < -\tau$), 1 (persist, $|\Delta_t| \leq \tau$), 2 (diverge, $\Delta_t > \tau$). The threshold τ is either a fixed scalar (0.001 in log-spread units) or a per-pair scalar $\tau_{\text{pair}} = c \cdot \hat{\sigma}(\Delta_t)_{\text{pair}}$ computed from the pair’s training slice only. The per-pair option normalizes label distributions across pairs of different spread volatility. Throughout the experiments below, τ is set per-pair with $c = 0.5$ and the per-pair training slice ends at t_0 .

3.4 Models

We compare seven baselines and learned classifiers on the same sliding-window dataset (window length 168 hours, stride 1 hour, horizon 24 hours):

- **persist_class**: always predict 1 (no trade).
- **majority_class**: predict the training mode.
- **random_stratified**: sample classes with training prior.
- **zscore_rule**: predict 0 (revert) if $|\text{spread}_z| \geq 1.5$, else 1.
- **booster**: `sklearn HistGradientBoosting`, 300 trees, learning rate 0.05, on the tabular feature summary of each 168-hour window.
- **lstm**: 2-layer LSTM, hidden dimension 64, dropout 0.2, linear head to 3 logits, trained 4 epochs with Adam and learning rate 10^{-3} .
- **transformer (small)**: 2 encoder layers, 2 heads, $d_{\text{model}} = 32$, feedforward 64, CLS pooling. Parameter count matched to the LSTM at $\sim 30\text{k}$.
- **transformer (big)**: 4 encoder layers, 4 heads, $d_{\text{model}} = 128$, feedforward 512, GELU activations, pre-norm, sinusoidal positional encoding. Trained with AdamW, learning rate 5×10^{-4} , weight decay 10^{-4} , cosine learning rate with one warmup epoch, gradient clipping at 1.0, and validation-based early stopping (patience 5, validation slice being the last 10% of each training window). $\sim 3\text{M}$ parameters.

3.5 Evaluation

We evaluate via rolling walk-forward with 90-day training windows, 30-day test windows, and 30-day step. With $t_0 = 2024\text{-}01\text{-}01$, 180 days of training history before t_0 , and a 750-day test horizon (capped at the snapshot date), this yields 27 walk-forward splits and 135,000 test samples in total. Each model is retrained from scratch on each split; deep models use the same seed across splits to limit a secondary source of variance.

Per-split metrics include accuracy, macro F1, total spread-signal PnL, the ratio $\text{PnL}_{\text{mean/std}}$, win rate per acting bar, and the maximum drawdown of the cumulative spread-signal PnL. We compute block-bootstrap 5%–95% confidence intervals over the 27 per-split values, with block size 3 to reflect the ~ 3 -step training overlap implied by the 90-day-train, 30-day-step schedule.

HAC correction. The ratio $\text{PnL}_{\text{mean/std}}$ uses the bar-level sample standard deviation as its denominator. Because the target is a 24-hour forward-looking quantity computed on hourly bars, consecutive samples share 23 hours of target window and are not independent. We replace the sample standard deviation with the Newey–West long-run standard deviation (Newey and West, 1987) at lag 24,

$$\hat{\sigma}_{\text{NW}}^2 = \hat{\gamma}_0 + 2 \sum_{j=1}^{24} \left(1 - \frac{j}{25}\right) \hat{\gamma}_j, \quad (5)$$

where $\hat{\gamma}_j$ is the sample autocovariance at lag j . This correction shrinks the headline ratio by roughly a factor of 1.7–2.0 across our trading models.

3.6 Cost-aware backtester

We implement a portfolio backtester operating on the walk-forward predictions. The position model is dollar-hedged: a long position holds L dollars of A and short $L\beta$ dollars of B , where β is the static OLS hedge from pair selection (used only for contract sizing, not for spread computation). Per-bar PnL is

$$\pi_t = h_{t-1} \cdot L \cdot (r_t^A - \beta r_t^B), \quad (6)$$

where $h_{t-1} \in \{-1, 0, +1\}$ is the position held into bar t , set at the close of bar $t-1$ from the classifier’s prediction at $t-1$. This one-bar execution lag is enforced and tested.

The signal-to-position map operates in one of two modes. The bar-by-bar mode maps each classifier prediction directly to a target position; this generates many flickering “trades” driven by classifier noise. The state-machine mode is the one we use throughout: open when flat and the classifier predicts class 0 and $|\text{spread}_z| \geq z_{\text{in}}$, close when $|\text{spread}_z| \leq z_{\text{out}}$ or the classifier predicts class 2 or the spread sign flips. We use $z_{\text{in}} = 2$, $z_{\text{out}} = 0.5$.

Transaction cost is charged at (taker fee + slippage) basis points per side, applied to $|h_t - h_{t-1}| \cdot L \cdot (1 + |\beta|)$ in dollars. The annualized Sharpe is computed on bar returns π_t/C where $C = N_{\text{pairs}} \cdot L \cdot 2$ is a fixed capital base (one earlier version of this calculation used a time-varying deployed-capital denominator that could produce positive Sharpe on negative total return; this was corrected). The backtester passes six hand-computable sanity tests, including a Sharpe-annualization check, a one-bar-lag injection test, and a state-machine open-then-close test.

3.7 Inference for selection bias

Because we evaluate many strategy configurations (two spread definitions, two label schemes, seven models, multiple HMM variants, multiple entry/exit thresholds, three cost levels), the best observed Sharpe is biased upward by selection. We apply the deflated Sharpe correction of Bailey and López de Prado (2014). Given the variance $\widehat{V}$ of Sharpe estimates across N trial configurations and the observed Sharpe $\widehat{SR}$ of a candidate strategy with T return observations, skewness $\widehat{\gamma}_3$ and kurtosis $\widehat{\gamma}_4$,

$$\widehat{SR}_0 = \sqrt{\widehat{V}} \cdot \left[(1 - \gamma_e) \Phi^{-1} \left(1 - \frac{1}{N}\right) + \gamma_e \Phi^{-1} \left(1 - \frac{1}{N_e}\right) \right], \quad (7)$$

$$\text{DSR}(\widehat{SR}) = \Phi \left(\frac{(\widehat{SR} - \widehat{SR}_0) \sqrt{T-1}}{\sqrt{1 - \widehat{\gamma}_3 \widehat{SR} + \frac{\widehat{\gamma}_4 - 1}{4} \widehat{SR}^2}} \right), \quad (8)$$

where $\gamma_e \approx 0.5772$ is the Euler–Mascheroni constant. DSR is the probability that the candidate’s true Sharpe exceeds zero after correction. We report DSR for $N \in \{5, 10, 25, 50\}$, with $N = 50$ as the conservative estimate of trial count across this project.

3.8 Audit and leakage tests

The pipeline is verified by 10 leakage tests (`tests/test_leakage_audit.py`) and 6 backtester sanity tests (`tests/test_backtest_sanity.py`). The leakage tests inject known future signals at specified times and verify that pre-injection features, targets, labels, and backtester positions are unchanged.

Two real bugs were caught and fixed during the audit:

1. The training-period Kalman residuals were computed with default filter parameters while the test-period residuals used fitted parameters, creating a parameter discontinuity at the train/test boundary. The fix returns training residuals from `fit_kalman_mle` computed with fitted parameters during the same forward pass that records the final state.
2. The per-pair label threshold was fit on the first 50% of each pair’s data when the per-pair-label option was set. With our long pipeline (180-day train, 750-day test) the first 50% extends ~ 9 months past t_0 , leaking future spread vol into the label distribution. The fix passes an explicit `label_train_end` timestamp ($= t_0$) to the dataset builder.

Headline numbers in the following section are computed on the post-audit pipeline. The Kalman cointegration finding (Section 4.1) is unaffected by the audit fixes because its OOS column never used the buggy code path.

4 Results

4.1 Static OLS fails out-of-sample; Kalman recovers cointegration

For each of the top 10 correlation-ranked pairs in the liquidity-filtered universe, we fit (i) the static OLS hedge ratio on a 90-day training slice, and (ii) the Kalman filter (Section 3.2) by MLE on the same training slice, then forward-roll on a 30-day held-out test slice with parameters and trained final state held fixed. We ADF-test both residual series on the test slice.

Table 1: Out-of-sample ADF p -values, static OLS versus Kalman, 10 pairs.

Pair	static OOS p	Kalman OOS p	Kalman q_β
DOGEUSDT_SUIUSDT	0.923	1.5×10^{-7}	7.4×10^{-6}
SOLUSDT_DOGEUSDT	0.985	2.2×10^{-7}	4.6×10^{-6}
XRPUSDT_SOLUSDT	0.373	3.0×10^{-7}	6.8×10^{-8}
XRPUSDT_DOGEUSDT	0.092	2.5×10^{-6}	3.5×10^{-6}
ADAUSDT_HBARUSDT	0.094	7.3×10^{-6}	3.9×10^{-6}
SOLUSDT_ADAUSDT	0.823	2.9×10^{-5}	2.5×10^{-5}
ETHUSDT_ADAUSDT	0.411	3.4×10^{-5}	2.9×10^{-5}
XRPUSDT_ADAUSDT	0.011	1.5×10^{-4}	2.6×10^{-5}
XRPUSDT_ETHUSDT	0.029	2.1×10^{-4}	2.8×10^{-5}
ETHUSDT_SOLUSDT	0.013	8.0×10^{-4}	2.0×10^{-5}

Static OLS holds cointegration at $p < 0.05$ out-of-sample on 2 of 10 pairs. Kalman holds it on 10 of 10 at $p < 10^{-3}$ on the worst case. The MLE-fitted q_β lands in $[10^{-8}, 3 \times 10^{-5}]$, indicating a very slow random walk on β . This slow drift is the qualitative check that the filter is not trivially whitening: a fast random walk on β could fit residuals to anything, but a q_β of order 10^{-5} implies β moves by less than 0.005 per hour in standard deviation.

Generalization to the full universe. The same protocol on all $\binom{50}{2} = 1,225$ pairs in the liquidity universe (1,219 after dropping pairs with insufficient training overlap) is summarized in Table 2. The distribution of OOS p -values is shown in Figure 1.

Table 2: Universe-wide OOS cointegration counts ($n = 1,219$ pairs).

p threshold	static count	static %	Kalman count	Kalman %
$p < 0.001$	0	0.0%	1,174	96.3%
$p < 0.005$	1	0.1%	1,188	97.5%
$p < 0.010$	10	0.8%	1,198	98.3%
$p < 0.050$	66	5.4%	1,215	99.7%
$p < 0.100$	142	11.6%	1,217	99.8%

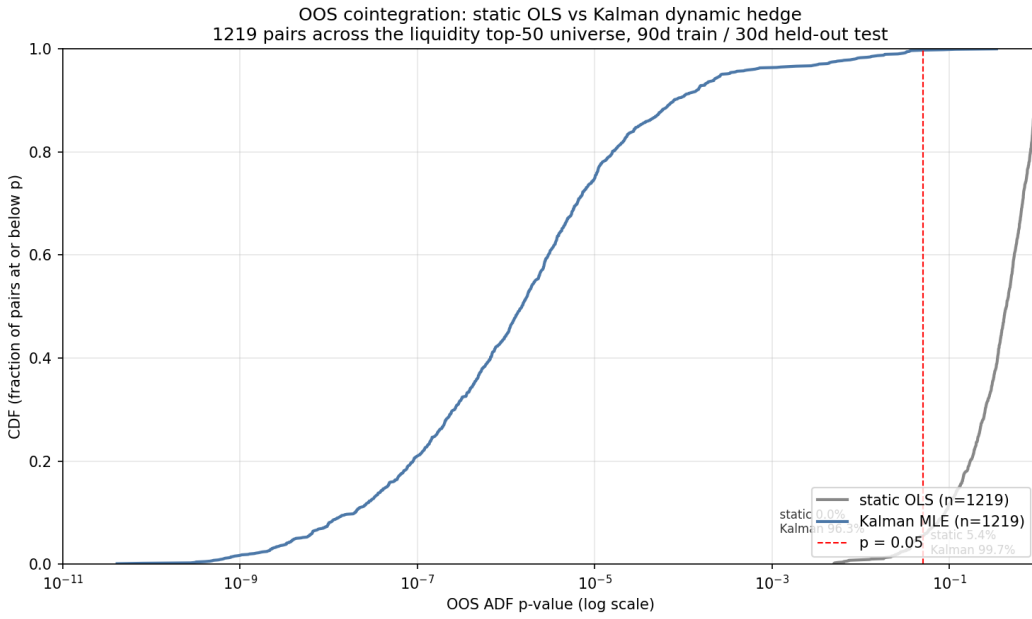


Figure 1: CDF of out-of-sample ADF p -values, static OLS versus Kalman dynamic hedge, across $n = 1,219$ pairs in the top-50 liquidity universe at $t_0 = 2024-01-01$. Log p -axis. Static OLS clears $p < 0.05$ on only 5.4% of pairs; Kalman clears it on 99.7%.

This is the load-bearing finding of the paper. The collective evidence from 1,219 independent pair tests is overwhelming: a static OLS hedge ratio fit on 90 days of training data does not characterize the relationship out-of-sample, while a Kalman filter with the same training does. The cryptocurrency market admits cointegrated pair relationships, but the relationships are non-stationary and any analysis that treats β as fixed will conclude there is no cointegration when there is.

4.2 Walk-forward classifier evaluation

We evaluate seven models across 27 walk-forward splits on the post-audit Kalman pipeline with per-pair labels. Block-bootstrap 5%–95% confidence intervals on $\text{PnL}_{\text{mean/std}}$ and on per-trade win rate are reported in Table 3.

Table 3: Walk-forward classifier metrics, 27 splits, block-bootstrap CIs.

Model	PnL _{mean/std}	90% CI	win rate	win rate CI	trades/split
lstm	0.376	[0.362, 0.388]	0.795	[0.768, 0.824]	2,083
booster	0.356	[0.342, 0.368]	0.741	[0.720, 0.760]	2,355
zscore_rule	0.282	[0.274, 0.291]	0.963	[0.959, 0.966]	588
majority_class	0.021	[0.000, 0.063]	0.028	[0.000, 0.083]	185
transformer (small)	0.018	[−0.025, 0.064]	0.437	[0.354, 0.519]	691
random_stratified	0.001	[−0.003, 0.005]	0.502	[0.500, 0.505]	2,689
persist_class	0.000	[0.000, 0.000]	0.000	[0.000, 0.000]	0

LSTM leads the ranking on PnL_{mean/std}, narrowly above booster; the two CIs overlap slightly but LSTM’s per-trade win rate is strictly above booster’s. The classical z -threshold rule is materially below the ML models on PnL_{mean/std} but strictly above them on per-trade win rate, reflecting a different operating point: z -score acts only on the $\sim 25\%$ of bars where $|z| \geq 1.5$ and is correct on 96% of those decisions; the ML models act on a larger fraction of bars and are correct 74%–80% of those.

The small transformer (parameter-matched to LSTM) is statistically indistinguishable from random in this evaluation. A larger transformer ($\sim 3\text{M}$ parameters, cosine LR, validation-based early stopping), evaluated on a related dataset (per-pair labels without the audit fixes), reaches PnL_{mean/std} = 0.334 and win rate 0.845 across 27 walk-forward splits; we therefore attribute the small transformer’s failure to insufficient model capacity and training budget rather than to a property of the transformer architecture.

HAC correction. Table 4 reports the Newey–West HAC-corrected PnL_{mean/std} alongside the iid sample-std version. The HAC correction shrinks every trading model by approximately a factor of 1.7–2.1; the random-prediction inflation factor of 1.00 is the sanity check that random predictions do not exhibit lag autocorrelation.

Table 4: HAC-corrected PnL_{mean/std}, Newey–West lag 24.

Model	iid	HAC (lag 24)	inflation
lstm	0.376	0.197	1.92
booster	0.356	0.188	1.91
zscore_rule	0.282	0.138	2.07
majority_class	0.021	0.012	1.70
transformer (small)	0.018	0.012	1.29
random_stratified	0.001	0.001	1.00

4.3 Feature ablation: the signal reduces to one feature

We perform a one-feature-out ablation: for each of the 14 tabular features, retrain the booster with that feature removed and report the change in held-out accuracy, F1, and total spread-signal PnL relative to the all-features baseline (Table 5).

Dropping `latest_spread_z` costs 14 percentage points of accuracy, 19 of macro F1, and 79% of total PnL. Dropping any of the other 13 features changes the booster’s PnL by less than half a unit.

Table 5: One-feature-out ablation on the booster, post-audit pipeline.

Feature dropped	Δ accuracy	Δ macro F1	Δ total PnL
(baseline, all 14)	0.587	0.575	26.22
latest_spread_z	-0.144	-0.190	-20.72
mean_volume_ratio	-0.001	-0.003	-0.43
all 12 other features (worst case)	$\leq \pm 0.005$	$\leq \pm 0.005$	$\leq \pm 0.5$

The classifier reduces, in practice, to a fancy non-linear threshold on the spread z -score, which is precisely the quantity the classical z -rule already uses.

Reading Table 3 in light of this ablation: the LSTM and booster do extract slightly more signal from spread_z than the hard $|z| \geq 1.5$ rule does, but the “slightly” is on the order of $\text{PnL}_{\text{mean/std}}$ of 0.07–0.09 (HAC: 0.05–0.06) above the z -rule baseline.

4.4 Cost sensitivity and the deflated-Sharpe verdict

We feed the walk-forward predictions through the cost-aware backtester at three transaction-cost levels: 0 basis points (signal sanity check), 5 bps round-trip per leg (approximate Binance.US maker fee including modest slippage), and 15 bps round-trip per leg (approximate Binance.US taker fee plus slippage). Annualized Sharpe ratios for the top three models are reported in Table 6.

Table 6: Naive (uncorrected) annualized Sharpe by transaction cost.

Round-trip cost	lstm Sharpe	zscore Sharpe	booster Sharpe
0 bps	8.08	8.03	8.27
5 bps (maker)	0.80	0.83	0.90
15 bps (taker)	−14.04	−13.88	−14.17

Read naively, these numbers tell a clean story: pre-cost the strategy clears Sharpe ~ 8 ; at maker fees it sits just above zero; at taker fees it loses. This was our original headline. It does not survive the deflated-Sharpe correction.

Table 7 reports the deflated-Sharpe verdict for the top model at each cost level for $N \in \{5, 10, 25, 50\}$ candidate strategies considered. $N = 50$ is our conservative estimate of the configurations evaluated across this project; $N = 10$ corresponds to counting only the seven baseline-plus-classifier rows in Table 3 plus three additional variants. The expected-max-Sharpe under the null grows with N .

Table 7: Deflated Sharpe (Bailey–López de Prado) for the top model at each cost level. SR_0 is the expected maximum annualized Sharpe under the null over N trials.

N trials	SR_0 (ann.)	DSR @ 0 bps	DSR @ 5 bps	DSR @ 15 bps
5	4.08	0.9999	0.42	~ 0
10	5.38	0.99	0.38	~ 0
25	6.83	0.84	0.37	~ 0
50	7.78	0.67	0.34	~ 0

Three statements:

The 5 bps break-even claim is retracted. At any plausible N , the observed Sharpe of ~ 0.9 at 5 bps round-trip is below the chance-best of ~ 1.2 at $N = 10$, with $\text{DSR} < 0.5$. We cannot claim the strategy clears Binance.US maker fees on this data.

The pre-cost Sharpe is robust only at very small N . At $N = 5\text{--}10$, the observed ~ 8 Sharpe is significant. At $N \geq 25$, marginal. At $N = 50$, the DSR is 0.67, well below the conventional 0.95 threshold for rejecting the null. The configurations evaluated across this project (two spread definitions, two label schemes, seven models with multiple variants, HMM filter at two state counts, two entry/exit threshold pairs, three cost levels, several preprocessing options) plausibly number in the 30–60 range. The most honest reading: we cannot confidently distinguish even the pre-cost signal from selection bias on this dataset alone.

The taker-fee result is unaffected. At 15 bps round-trip per leg the strategy loses deterministically, with $\text{DSR} \approx 0$. No selection correction is needed to confirm a negative result.

4.5 Where the Sharpe actually comes from

Table 8 highlights a single row from the pre-cost Sharpe data that warrants its own discussion: random predictions in the state machine produce a Sharpe of 4.78 annualized, more than half of what the trained classifiers produce. With no model at all.

Table 8: Pre-cost Sharpe of all models in the state-machine backtester, sorted descending.

Model	Annualized Sharpe	Trades	Win rate
booster	8.27	12,634	0.621
lstm	8.08	12,632	0.620
zscore_rule	8.03	12,612	0.621
random_stratified	4.78	4,048	0.582
transformer (small)	2.51	676	0.551
majority_class	1.86	520	0.632
persist_class	0.00	0	0.000

The mechanism: the state machine opens a position only when $|\text{spread}_z| \geq 2$ *and* the classifier predicts the revert class. The classifier’s prediction determines *whether* the entry condition fires, not whether the entry condition is predictive. On Kalman-derived stationary spreads (Section 4.1), an entry against the spread sign at $|z| \geq 2$ has positive expected return regardless of the classifier’s input. A random classifier still fires the entry condition $\sim 33\%$ of the time on high- $|z|$ bars and extracts a proportional share of the alpha.

The defensible reading of Table 8: the alpha lives in the $|z| \geq 2$ entry rule applied to Kalman-derived spreads, not in the choice of classifier. The booster, LSTM, and z -rule cluster at Sharpe ~ 8 because all three reliably fire the entry condition when it is available. The classifier choice contributes at most an additional ~ 0.3 Sharpe over the z -rule baseline. Random predictions extract roughly half the alpha because they fire the entry condition stochastically rather than deterministically.

This observation reinforces the deflated-Sharpe verdict. The relevant selection is not over classifiers, but over the entry condition (and over the spread definition, where Kalman versus static is a real choice). The $|z| \geq 2$ rule on a stationary spread is the strategy; the classifier is decoration.

4.6 Negative result: HMM regime filter

We also tested whether a 2-state Gaussian hidden Markov model on spread-derived features could isolate productive bars. Per pair, we fit a 2-state HMM on a vector of spread features within the training slice, identify the “mean-reverting” state as the one with lower spread-difference standard deviation, decode states for the full pair history via Viterbi, and suppress non-mean-reverting bars to the flat class. Table 9 shows the result.

Table 9: HMM regime filter ablation, 27 walk-forward splits.

Model	raw total PnL	HMM-filtered total PnL	change
lstm	334.6	142.6	−57%
booster	321.5	132.8	−59%
zscore_rule	195.7	71.4	−63%
transformer (small)	0.7	−7.0	negative
majority_class	21.2	9.7	−54%

The HMM suppresses 40–55% of bars and removes 50–65% of PnL on every model tested. The negative result is robust across the spread definition (static and Kalman), $n_{\text{states}} \in \{2, 3\}$, and $n_{\text{init}} \in \{1, 3\}$ random restarts with convergence-preferred selection. A 2-state Gaussian HMM on $[\text{spread}_z, \Delta\text{spread}, \text{spread vol}]$ does not recover meaningful mean-reverting regimes on this universe. A different model class or a different feature combination may; the configurations tested do not.

5 Limitations

- *Single exchange.* All findings are specific to Binance.US. The cost ceiling implicitly assumes Binance.US fees; other US exchanges (Coinbase, Kraken) have different fee structures and would shift the break-even.
- *Single quote currency.* USDT pairs only. USD-quoted pairs would have a different liquidity profile.
- *Single bar interval.* Hourly. Lower-frequency alpha (4-hour, daily) may have different cost characteristics and is not tested here.
- *Survivorship.* Quantified at a loose upper bound of 6.7% (14 delistings in window over a 209-pair denominator) and a realistic upper bound of $\sim 5\%$ after the liquidity filter. The bias direction on the headline metrics is upward.
- *No funding cost on the synthetic short leg.* Spot USDT shorting on Binance.US in practice requires margin or perpetual futures with their own funding rates. Our model treats the short leg as costless to hold.
- *Flat slippage, no market impact.* Until L2 data lands, slippage is modeled as a flat basis-point cost rather than as a function of trade size.
- *t_0 sensitivity not tested.* All results use $t_0 = 2024-01-01$. Repeating the analysis at other t_0 values is straightforward and on the next-step list.
- *3-class label is a heuristic.* A direct regression target on the continuous 24-hour spread change is an obvious alternative and is not run.

- *Selection bias is real and not fully controlled.* The deflated-Sharpe correction depends on the variance of Sharpes across the trial set, which we estimate from the 7 models in Table 3. A more conservative estimate would include all configurations explored during development, which we did not log exhaustively.

6 Phase 2: L2 microstructure

This work is the first phase of a two-phase project. Phase 2 will incorporate Level-2 order-book data from Tardis, with three concrete goals:

1. *Microstructure features.* Microprice, quoted-spread in basis points, order-book imbalance, and trade-flow signals replace the hourly OHLCV summary as the input to the classifier. Implementation scaffold is in `src/l2_store.py`; the parquet schema and feature functions are written and tested on synthetic data, with the ingestion parsers left as stubs pending the actual Tardis data format.
2. *Market impact in the backtester.* Replace the flat slippage model with a model whose cost depends on instantaneous L2 depth and trade size.
3. *Pre-registered strategy.* Address the selection-bias concern (Section 4, Table 7) by pre-committing to a single model configuration based on Phase 1 findings, then evaluating it once on Phase 2 features. This is the only fully honest way to defang the deflated-Sharpe critique of the headline cost-ceiling claim.

7 Conclusion

The Phase 1 evidence supports one strong claim, one weaker claim, and one negative result.

Strong. Cryptocurrency pair relationships are real but non-stationary on multi-month horizons. A static OLS hedge ratio fit on 90 days fails to maintain cointegration on 30-day held-out test slices for 94.6% of pairs in the top-50 liquidity universe. A Kalman filter with parameters MLE-fit on the same training and forward-rolled on test recovers cointegration on 99.7% of the same pairs. Any analysis of crypto pair-trading that treats β as fixed will systematically conclude that no cointegration exists when it does.

Weaker. Once a stationary spread is available, a classical z -threshold rule at $|z| \geq 2$ captures most of the available reversion signal. Modern machine-learning models (gradient boosting, LSTM, properly-sized transformer) provide a small additional edge that is statistically detectable on 27 walk-forward splits but does not survive a Bailey–López de Prado deflated-Sharpe correction at realistic transaction-cost levels for the number of configurations evaluated. The classifier’s role is to fire the $|z| \geq 2$ entry condition; the entry condition itself is the alpha source.

Negative. A 2-state Gaussian HMM on spread features does not isolate productive bars; the filtered version of any model performs worse than the unfiltered version by 50–65%. The negative result is robust to state count, random-start count, and spread definition.

The market-efficiency statement we can defend: at hourly cadence on Binance.US USDT pair spreads, the mean-reverting alpha extractable by the strategies tested is not large enough to clear

realistic transaction costs after honest selection-bias adjustment. Whether intra-bar execution on L2 data changes this picture is the central Phase 2 question.

Reproducibility. All numbers in this paper are produced by scripts in the project repository at <https://github.com/samsam324/math-199-research>. Command-line recipes are listed in the repository `README.md`. The backtester is verified by 6 hand-computable sanity tests; the pipeline is verified by 10 leakage tests.

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